

YULIANG FU

North Carolina State University ◇ Raleigh, NC, USA

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EDUCATION

North Carolina State University, Raleigh, USA

Aug, 2024 – May, 2029 (Expected)

PhD in Computer Science (*In Progress*)

Tsinghua University, Beijing, China

Aug, 2020 – Jul, 2024

BS in Mathematics and Physics + **BS** in Electrical Engineering and Automation

RESEARCH INTERESTS

Human-Computer Interaction, Mobile Computing, Biomedical Computing, and AIoT

RESEARCH EXPERIENCE

SSCS Lab, North Carolina State University, Raleigh, USA

Department of Computer Science

Aug, 2024 – Present

PhD Student

Advisor: Dr. [Chenhan Xu](#)

- Explore wireless fingertip force sensing for interactive applications and disease detection
- Develop novel input protocols for smart wearable devices

SciFi Lab, Cornell University, Ithaca, USA

College of Computing and Information Science

Jun, 2023 – Sep, 2023

Student Research Intern

Advisor: Dr. [Cheng Zhang](#)

- Explored the possibility of commodity headsets in acoustic sensing
- Developed a facial expression tracking and movement recognition algorithm with FMCW signal

Pervasive HCI Group, Tsinghua University, Beijing, China

Department of Computer Science and Technology

Apr, 2022 – Jun, 2023

Research Assistant

Advisor: Dr. [Yuntao Wang](#)

- Developed an acoustic ranging method for home AIoT devices
- Proposed an indoor location service utilizing ultrasonic waves

PROFESSIONAL EXPERIENCE

Research and Development Intern, JD Logistics, Beijing, China

Department of Technology and Data Intelligence

Nov, 2023 – Jan, 2024

Mentor: Dr. [Desheng Zhang](#)

- Performed data mining in the scene of Last-Mile Delivery
- Analyzed couriers' workload based on IMU data from terminal devices

PUBLICATIONS

- [3] **Yuliang Fu**, Zhi Zhang, Rakshita Ranganath, Zhizhen Li, Yuchen Liu, Ning Sui, Huining Li, Chenhan Xu, “*TouchWave: Exploring mmWave-based Non-contact Fingertip-force Sensing in Activities of Daily Living*”, IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025), Los Angeles, CA, Nov 2025 (To appear)
- [2] Zhi Zhang, Chongxin Zhong, **Yuliang Fu**, Ying Chen, Jinyuan Jia, Huining Li, Chenhan Xu, “*mV-IMU: mmWave-enabled Virtual Inertia Measurement Unit for High-fidelity Activities of Daily Living Monitoring*”, IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025), Los Angeles, CA, Nov 2025 (To appear)
- [1] Rakshita Ranganath, Zhi Zhang, **Yuliang Fu**, Jinyuan Jia, Ying Chen, Huining Li, Chenhan Xu, “*Hand-Grip Strength Estimation through Bioacoustic Sensing*”, IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025), Los Angeles, CA, Nov 2025 (To appear)

LEADERSHIP & SERVICE

Secretary of Students’ Science Association, Weiyang College, Tsinghua University

Oct, 2020 – Jan, 2023

- Organized 2020/2021/2022 Weiyang College freshman technology innovation competition, attracting more than 50 participants each year

SCHOLARSHIPS & AWARDS

- 2021 Tsinghua Comprehensive Excellence Award
- 2021 Tsinghua Technology Innovation Award

TECHNICAL SKILLS

Programming	Python, C/C++, SQL, MATLAB
Embedded System	Nordic nRF52 series, TI MSP430, STM32 series, Arduino
Signal Processing	mmWave Radar, Ultrasonic Sensing
Others	Verilog HDL, AutoCAD, Solidworks